MIL-G-46795F(AT)
31 August 1981
SUPERSEDING
MIL-G-46795E(AT)
23 September 1975

MILITARY SPECIFICATION

GENERATING SYSTEM: ALTERNATOR, RECTIFIER 60 OR 25 AMPERES, 28 VOLTS

This specification is approved for use by US Army Tank-Automotive Command, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers 28 Volt, direct current (Vdc) generating systems, engine driven through belt or direct connection, to provide 60 or 25 amperes for charging and operating loads. System consists of a rotating field alternator with integral diode rectifier and adjustable voltage regulator in a single housing. This specification also covers repair parts for this system.

1.2 Classification. (see 6.4).

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-B-11188 - Batteries, Storage; Lead-acid.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: US Army Tank-Automotive Command, ATTN: DRSTA-GSS, Warren, MI 48090, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document, or by letter.

FSC 2920

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STANDARDS

FEDERAL

FED-STD-H28/2	Screw Thread Standards for Federal Services.

MILITARY

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-130	Identification Marking of US Military Property.
MIL-STD-202	Test Methods for Electronic and Electrical Component Parts.
MIL-STD-461	Electromagnetic Emission and Susceptibility Requirements for the Control of Electromagnetic Interference.
MIL-STD-462	Electromagnetic Interference Characteristics, Measurement of.
MIL-STD-810	Environmental Test Methods.
MIL-STD-889	Dissimilar Metals.
MIL-STD-1275	Electrical Circuit, 28 Volts, DC Voltage Characteristics in Military Vehicles.

DRAWINGS

ARMY

10929868	Rectified AC Generating System.

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity, or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 First article.

3.1.1 Generating system. The contractor shall furnish sample units for first article inspection and approval (see 4.4 and 6.3). First article samples shall be inspected by the contractor under the surveillance of the Government to determine conformance to quality assurance provisions of this specification. First article samples shall be fully representative of generating systems to be supplied from production tooling and facilities.

3.1.2 Repair parts. Individual repair parts furnished for first article approval shall be substituted in a previously approved generating system. Generating system, with repair part functioning as in Intended use, shall Subsequently undergo and pass first article inspection as specified in 3.1.1.

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3.2 Materials. Materials shall be as specified herein and in referenced specifications, standards, and drawings. Material shall be free of defects which adversely affect performance or serviceability of the finished product (see 6.5).

3.2.1 Dissimilar metals. Dissimilar metals shall be protected from galvanic corrosion in accordance with requirements of MIL-STD-889.

3.3 Design and construction. Generating system shall be constructed to the form and dimensions specified on drawing 10929868, and applicable specifications and standards.

3.3.1 Interchangeability. Generating system and all component parts shall be physically and functionally interchangeable with all other such parts of the same part number previously furnished under this specification.

3.3.2 Electromagnetic interference characteristics. Generating systems shall conform to requirements specified for class A3 of MIL-STD-461.

3.3.3 Polarity. Generating systems shall be designed to operate in negatively grounded electrical circuit.

3.3.4 Standard parts. Military standard parts shall be incorporated whenever possible. Commercial standard parts may be used, provided they are interchangeable with military standard parts without modification.

3.3.5 Threaded parts, Screw threads of the form, number per inch, and class specified on the applicable drawing or military standard shall be in accordance with FED-STD-H28/2.

3.4 Performance.

3.4.1 Voltage. Generating system shall provide steady state (see MIL-STD-1275) voltage at any setting between 26 Vdc and 30 Vdc when mechanically and electrically installed and operated as in intended use.

3.4.1.1 Voltage adjustment. Generating system shall provide means to adjust voltage output between 26 Vdc and 30 Vdc.

3.4.1.2 Voltage regulation. Generating system, when output voltage is adjusted as in intended use, shall maintain set voltage (see 6.6.1) within + 0.5 Vdc . Further, generating system shall return to set voltage within 0.6 second of any load change.

3.4.1.3 Transient voltage characteristics. Generating system shall conform to requirements of MIL-STD-1275 regarding ripple, surges, spikes, and starting disturbances in fault-free, single fault, and multi-fault conditions.

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3.4.2 Current (output). Generating system shall be adjustable to one of two fixed upper limits of current (output): maximum (rated output), or reduced (reduced output).

3.4.2.1 Maximum current (rated output). Generating system shall supply 60 amperes (amp) current under maximum load at 10,000 revolutions per minute (rpm) alternator rotor shaft speed.

3.4.2.2 Reduced current (reduced output). When properly adjusted, generating system shall limit output to 25 amp current under maximum load at 10,000 rpm alternator rotor shaft speed.

3.4.2.3 Non-operating current. Generating system shall exhibit non-operating current through ignition switch circuit of not more than 300 milliamperes. Further, with ignition switch open, non-operating current from battery shall be not more than 25 milliamperes.

3.4.3 Endurance. Generating system shall maintain set voltage (see 3.4.1.2) during and subsequent to 3000 hours of operation under various conditions of rotor shaft position, speed, and current load as specified in table I. There shall be no servicing, adjustment, nor replacement of parts.

TABLE I. Endurance requirements.

Load current	Rotor shaft speed (rpm)	Time at Speeds (hours)	Rotor shaft position
60	2500	1008	Horizontal
60	10000	498	Horizontal
1/	6000	498	Horizontal
60	6000	498	45° ± 1° above horizontal
60	6000	498	45° ± 1° below horizontal

1/ Load current shall be cycled from minimum to maximum during this portion of schedule.

3.4.4 Overspeed. Generating system shall operate at 12,000 rpm under not more than 5 ampere load for 15 seconds without impairing electrical or mechanical performance.

3.4.5 High temperature operation. Generating system shall maintain set voltage output ± 0.2 Vdc during 200 hours of operation in ambient air temperature of 250°F. Subsequently, system shall conform to 3.4.2.

3.4.6 Low temperature operation. Generating system shall maintain set voltage output ± 0.5 Vdc during four hours of operation in ambient air temperature of -65°F after 24 hours of storage in ambient air temperature of -80°F. Subsequently, system shall conform to 3.4.2.

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3.4.7 Shock. Generating system shall undergo test procedure specified in condition I, method 213, shock test of MIL-STD-202. Subsequently, system shall conform to 3.4.1 and 3.4.2.

3.4.8 Vibration. Generating system shall maintain set voltage Output + 0.5 Vdc while undergoing test procedure specified in condition D, method 204 zf MIL-STD-202.

3.4.9 Submersion. Generating system shall operate during eight hours of submersion in a salt-water solution. Subsequently, voltage regulator compartment shall contain not more than 10 cubic centimeters of salt-water solution. Further, system shall conform to 3.4.1 and 3.4.2.

3.4.10 Fungus resistance. Generating system shall undergo test procedure specified in method 508 of MIL-STD-810 for test period of 90 days. Subsequently, system shall conform to 3.4.1 and 3.4.2.

3.4.11 Corrosion resistance. Generating system shall undergo test procedure specified in method 101 of MIL-STD-202 for test period of 200 hours. Subsequently, system shall conform to 3.4.1 and 3.4.2.

3.4.12 Sand and dust. Generating system shall undergo test procedure specified in method 510, sand and dust test of MIL-STD-810. Subsequently, system shall conform to 3.4.1 and 3.4.2.

3.5 Finish. Generating system shall be painted as specified in applicable drawings, specifications, and standards on all exterior surfaces, except electrical connections and exposed shaft surfaces.

3.6 Identification of product.

3.6.1 Marking. Generating system shall be marked in accordance with MIL-STD-130.

3.6.2 Nameplate. Generating system shall have a nameplate attached to alternator frame in conspicuous location so that plate may be easily read when generating system is installed in vehicle. Nameplate shall contain following minimum information:

GENERATING SYSTEM DC
60 AMPS/25 AMPS 28 VOLTS
MAXIMUM RATED SPEED - 10,000 RPM
SPECIFICATION MIL-G-46795
SERIAL NUMBER
NATIONAL STOCK NUMBER (NSN)
MANUFACTUER'S NAME OR IDENTIFICATION CODE
us

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3.7 Workmanship. Workmanship shall be such as to assure a product free of burrs, scratches, sharp edges, chips, corrosion, scale, and dirt.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Materials. Contractor's inspection records shall be reviewed to determine conformance to 3.2, 3.2.1, 3.3.4, and 3.3.5.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.4).
b. Quality conformance inspection (see 4.5).

4.3 Inspection conditions. Unless otherwise specified herein, all examinations and tests shall be performed under conditions specified in 4.3.1 through 4.3.4.

4.3.1 Apparatus. Unless otherwise specified herein, inspection apparatus shall include wiring, switches, meters, and direct current power sources arranged to include circuitry as specified in figures 1 and 2. Apparatus shall also include the following:

a. Chamber(s) for controlling ambient air temperature between minus 80°F and plus 260°F, and for conducting sand and dust tests.
b. Tank(s) in which to submerge generator system in salt-water solution.
c. Temperature indicating equipment suitable to disclose ambient air temperatures and generating system frame temperatures.
d. Mechanism to drive, and to control driven speed, of generating system up to 1.2,000 rpm.
e. Provisions for operating generating system while in temperature or sand and dust chamber, or while submerged, and whether connected or disconnected in test circuitry.
f* Tachometer.
g. Prony brake measuring instrument.
h. Milliammeter, 500 milliamp full scale.

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i. Ammeter DC, 75 amp full scale.
j. Voltmeter, DC, 30 volts full scale.

Apparatus shall permit measurement within 10 percent of tolerance specified for product or test with gages and meters accurate as specified in 4.3.1.1 and 4.3.1.2.

4.3.1.1 Gage and meter accuracy. Unless otherwise specified herein, gages and meters shall be accurate within tolerances specified as follows:

Gage or meter	Accuracy tolerance percent of scale reading
Ammeter, 75 amp full scale	± 0.50%
Milliammeter, 500 milliamps full scale	± 0.50%
Prony brake	± 2.0%
Tachometer	± 1.0%
Temperature indicating system	± 8 degrees F
Voltmeter, 30 volts full scale	± 0.25%

4.3.1.2 Gage and meter calibration. Unless otherwise specified herein, calibration of test equipment shall be conducted at intervals sufficient to maintain accuracy as specified in 4.3.1.1, or 4.5.3.4. Records of calibration shall be made available to the Government. The Government may refuse to conduct an inspection where accuracy of gages and meters is not satisfactorily supported by calibration records.

4.3.1.3 Batteries. 24 volt power source shall be two 12 volt batteries connected in series. Batteries shall conform to MIL-B-11188. A battery which accepts a charging current of not more than 4.5 amps at set voltage after storage at laboratory ambient air temperature shall be considered fully charged. In case of a high voltage reading, battery fluid specific gravity shall be inspected and, if over 1.30, battery shall be discharged until specific gravity is not more than 1.30, and charging current is not more than 4.5 amperes.

4.3.2 Stabilized frame temperature. Unless otherwise specified herein, generating system frame temperature shall be stabilized prior to undergoing test procedure.

4.3.2.1 Stabilized temperature. Frame temperature shall be considered stabilized when it varies not more than 8°F between high and low readings in any one hour of operation under constant conditions.

4.3.2.2 Frame temperature. Frame temperature shall be measured at center of stator lamination stack on outside diameter.

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4.3.3 Ambient air temperature. Unless otherwise specified herein, room ambient air temperature shall be maintained at 77°F ± 15°F. Tests shall be conducted under general laboratory conditions acceptable to the Government. Specimens shall be stabilized for one hour before being subjected to tests. Unless otherwise specified, specimens shall remain at set voltage during all tests.

4.3.4 Test voltage. Unless otherwise specified, voltage regulator shall be adjusted to 28 Vdc ± 0.2 Vdc with alternator running at 2000 rpm and producing not more than 10 amps. This adjustment shall be made after frame temperature has been stabilized.

4.4 First article inspection. Samples of products to be furnished under this specification shall undergo examinations in 4.5.3.2 and tests in order listed in table 11.

4.4.1 Reduced level first article inspection. Reduced level first article inspection may be performed if approval has first been granted (see 6.3.1). Samples of products to be furnished under this specification shall undergo examinations in 4.5.3.2 and tests in order listed in table II, except 4.6.4.1 shall be substituted in place of 4.6.4.

4.4.2 Generating system. Seven generating systems (see 3.1.1) shall be identified for first article inspection and shall undergo examinations and tests as specified in 4.4 (see 4.1, 6.2, and 6.3).

4.4.3 Repair parts. Seven samples of an individual part (see 3.1.2) shall be substituted as in intended use in generating systems previously approved as having undergone first article inspection (see 4.4). Subsequently, generating systems shall undergo examinations and tests specified in 4.4 (see 4.1, 6.2 and 6.3).

TABLE II. Order of first article testing.

Sample number	Requirement	Test	Requirement
1	3.4.2.1	4.6.3.1	Rated output
	3.4.1.2	4.6.2.2	Voltage regulation
	3.4.2.2	4.6.3.2	Reduced output
	3.4.2.3	4.6.3.3	Non-operating current
8/	3.4.3	4.6.4	Endurance
	3.4.2.1	4.6.3.1	Rated output
	3.4.1.2	4.6.2.2	Voltage regulation
2	3.4.2.1	4.6.3.1	Rated output
	3.4.1.2	4.6.2.2	Voltage regulation
1/	3.4.5	4.6.6	High temperature

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TABLE II. Order of first article testing. - Continued

Sample number	Requirement	Test	Requirement
1/	3.4.1.2	4.6.2.2	Voltage regulation
2	3.4.2.1	4.6.3.1	Rated output
2/	3.4.6	4.6.7	Low temperature
2/	3.4.1.2	4.6.2.2	Voltage regulation
	3.4.2.1	4.6.3.1	Rated output
	3.4.7	4.6.8	Shock
3/	3.4.2.1	4.6.3.1	Rated output
3/	3.4.1.2	4.6.2.2	Voltage regulation
3	3.4.2.1	4.6.3.1	Rated output
	3.4.1.2	4.6.2.2	Voltage regulation
	3.4.4	4.6.5	Overspeed
	3.4.2.1	4.6.3.1	Rated output
4/	3.4.8	4.6.9	Vibration
4/	3.4.1.2	4.6.2.2	Voltage regulation
	3.4.2.1	4.6.3.1	Rated output
	3.4.1.2	4.6.2.2	Voltage regulation
	3.4.12	4.6.13	Sand and dust
	3.4.2.1	4.6.3.1	Rated output
	3.4.1.2	4.6.2.2	Voltage regulation
4	3.4.2.1	4.6.3.1	Rated output
	3.4.1.2	4.6.2.2	Voltage regulation
	3.4.9	4.6.10	Submersion
5/	3.4.2.1	4.6.3.1	Rated output
	3.4.1.2	4.6.2.2	Voltage regulation
5	3.4.2.1	4.6.3.1	Rated output
	3.4.1.2	4.6.2.2	Voltage regulation
	3.4.1.3	4.6.2.3.2	Load switching
	3.4.2.1	4.6.3.1	Rated output
	3.4.1.3	4.6.2.3.1	High voltage transients
	3.4.2.1	4.6.3.1	Rated output
6	3.4.11	4.6.12	Corrosion resistance (200 hours)
6/	3.4.2.1	4.6.3.1	Rated output
6/	3.4.1.2	4.6.2.2	Voltage regulation
7	3.4.10	4.6.11	Fungus resistance (90 days)
7/	3.4.2.1	4.6.3.1	Rated output
7/	3.4.1.2	4.6.2.2	Voltage regulation

1/ 2/ 3/ 4/ 5/ 6/ and 7/ Samples shall undergo simultaneously those tests marked with same footnote number.

8/ 4.6.4.1 shall be substituted for 4.6.4 only when reduced level first article testing is approved (see 6.3.1).

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4.5 Quality conformance inspection.

4.5.1 Sampling - generator systems.

4.5.1.1 Lot formation. A lot shall consist of all generating systems, of one type or part number, from an identifiable production period, from one manufacturer, submitted at one time for quality conformance inspection.

4.5.1.2 Sampling for examination. Samples for quality conformance examination shall be selected in accordance with MIL-STD-105.

4.5.1.3 Sampling for acceptance testing. All generating systems (100%) furnished under this specification shall undergo acceptance tests (see 4.5.5).

4.5.2 Sampling - repair parts.

4.5.2.1 Lot formation. A lot shall consist of all repair parts, of one type or part number, from an identifiable production period, from one manufacturer, submitted at one time for quality conformance inspection.

4.5.2.2 Sampling for examination. Samples for quality conformance shall be selected in accordance with MIL-STD-105.

4.5.2.3 Sampling for acceptance testing. All repair parts (100%) furnished under this specification shall undergo acceptance tests (see 4.5.5).

4.5.3 Examinations.

4.5.3.1 Acceptable quality level. Generating systems selected in accordance with 4.5.1.2, or repair parts selected in accordance with 4.5.2.2, shall be examined to determine conformance to the following Acceptable Quality Levels (AQL's), on the basis of percent defective:

Classification	AQL
Major	1.0
Minor	2.5

4.5.3.2 Classification of defects. For examination purposes, defects shall be classified as listed in table 111.

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TABLE III. Classification of defects.

Category	Defect	Method of inspection
Major		
101	Dimensions affecting interchangeability not within tolerance (see 3.3 and 3.3.1)	SIE
102	Improper polarity (see 3.3.3)	Meter
Minor		
201	Dimensions not affecting interchangeability not within tolerance (see 3.3 and 3.3.1)	SIE
202	Improper finish (see 3.5)	Visual
203	Improper marking (see 3.6)	Visual
204	Poor workmanship (see 3.7)	Visual

SIE = Standard Inspection Equipment.

4.5.3.3 Gage and meter accuracy - acceptance and control testing. Unless otherwise specified herein, gages and meters used during acceptance testing and control testing shall be accurate within tolerances specified as follows:

Gage or meter	Accuracy tolerance plus or minus
Voltmeter DC	0.15 Vdc
Ammeter	0.5 amps
Tachometer	50 rpm at 1800 rpm
Tachometer	200 rpm at 10000 rpm
Temperature indicating system	8°F
Milliammeter	5.0 milliamps

4.5.3.4 Gage and meter calibration - acceptance and control testing. Calibration of gages and meters specified in 4.5.3.3 shall conform to 4.3.1.2.

4.5.4 Classification of tests. Test requirements specified herein are classified as follows:

a. Acceptance tests (see 4.5.5).
b. Control tests (see 4.5.6).

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4.5.5 Acceptance tests. All generating systems (100%), or repair parts (100%), furnished under this specification shall be subjected to acceptance tests specified in table IV, in order listed. Stabilized frame temperature will not be required.

TABLE IV. Order of acceptance tests.

Test	Requirement	Test
Rated output	3.4.2.1	4.6.3.1.1
Voltage regulation	3.4.1.2	4.6.2.2.1
Non-operating current	3.4.2.3	4.6.3.3

4.5.5.1 Failure - generator systems. Generating systems that have been rejected as the result of testing specified in 4.5.5, may be reworked, or have parts replaced, to correct defects, and be resubmitted for examination and tests. Before resubmitting, detailed information concerning rejection and action taken to correct defects shall be furnished to the Government.

4.5.5.2 Failure - repair parts. Repair parts, substituted in previously approved generating system for testing purposes, shall be considered as having failed a test when generating system fails that test. Failure of acceptance test sample to pass any specified examination or test may be cause for the Government to refuse to accept subsequent lots and to return lots previously accepted which can be identified with failed sample. Contractor must prove to satisfaction of the Government that any deficiency revealed by examination or test has been corrected.

4.5.6 Control tests.

4.5.6.1 Sampling. Control test samples shall be selected from a production lot which has passed quality conformance examination as specified in 4.5.3 and acceptance tests as specified in 4.5.5. Control test samples shall be selected at the rate of one generator system for each 250 produced except that not more than two shall be selected in any 30 day period.

4.5.6.2 Procedure. Control test samples shall be identified as to production period and subjected to tests specified in table V.

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TABLE V. Order of control tests.

Test	Requirement	Test
Reduced output	3.4.2.2	4.6.3.2 (2500 rpm only)
Overspeed	3.4.4	4.6.5
Voltage regulation	3.4.1.2	4.6.2.2
Submersion	3.4.9	4.6.10
Transient voltage	3.4.1.3	4.6.2.3 at 77°F ± 15°F only

4.5.6.3 Failure - generator systems. Failure of control test sample to pass any specified examination or-test may be cause for the Government to refuse to accept subsequent lots and to return lots previously accepted which can be identified with failed sample. Contractors must prove to satisfaction of the Government that any deficiency revealed by examination or test has been corrected .

4.5.6.4 Failure - repair parts. Repair parts substituted in previously approved generating system for testing purposes shall be considered as having failed test when generating system fails that test. Failure of control test sample to pass any specified examination or test may be cause for the Government to refuse to accept subsequent lots, and to return lots previously accepted which can be identified with failed sample. Contractor must prove to satisfaction of the Government that any deficiency revealed by examination or test has been corrected.

4.6 Conformance verification.

4.6.1 Electromagnetic interference characteristics test. To determine conformance to 3.3.2, generating system shall be subjected to tests specified in MIL-STD-462.

4.6.2 Voltage test. To determine conformance to 3.4.1, generating system shall undergo procedures specified in 4.6.2.1 through 4.6.2.3.2.

4.6.2.1 Voltage adjustment test. To determine conformance to 3.4.1.1, generating system shall be connected in circuit as shown in figure 1 and subjected to following procedure:

a. At generator speed of 2400 rpm, using battery load only, adjust regulator potentiometer to maximum and minimum and record voltage.
b. Increase load to 60 amperes, repeat adjustments, and record voltage as in a.
c. Increase generator speed to 8000 rpm and repeat a and b.

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4.6.2.2 Voltage regulation test. To determine conformance to 3.4.1.2, voltages shall be recorded with generating system operating under following conditions:

	RPM	AMPs
a.	2500 ± 50	0-4 . 5
	2500 ± 50	60
	2500 ± 50	0-4 . 5
	2500± 50	30
	2500 ± 50	0-4 . 5
	2500 ± 50	45
b.	6000 ± 120	Same sequence as in a.
C.	10000 ± 200	Same as in a.

NOTE : Change in load resistance shall be accomplished in 10 milliseconds maximum time.

4.6.2.2.1 Voltage regulation - acceptance test. All generating systems furnished under this specification shall be operated in ambient air temperature of 77°F ± 15°F with frame temperature non-stabilized through the following cycle of speeds and loads:

	RPM	AMPs
a.	2500 ± 50	0-4.5
b.	2500 ± 50	60
c.	10000 ± 200	0-4.5
d.	10000 ± 200	60

During this procedure , generating system shall conform to 3.4.1.2

4.6.2.3 Transient voltage characteristics test. To determine conformance to 3.4.1.3, generating system shall be subjected to test procedures specified in MIL-STD-1275. Further, system shall be subjected to test procedures specified in 4.6.2.3.1 and 4.6.2.3.2.

4.6.2.3.1 High voltage transients. Generating system shall be connected in test circuit as shown in figure 1 with SW2 and SW3 open and SW1 closed at zero rpm. A positive voltage pulse of 100 volts square wave of 50 milliseconds duration shall be applied to generating system output terminal once every two seconds for a period of 10 minutes. Generating system shall then be connected in a test circuit as shown in figure 2 with SW1 in position 1. Capacitator C shall be charged from power source, through R, until V1 reads 250 + 5 volts. SWI shall then be changed from position 1 to position 2 within one second and held in that position for 15 seconds. This procedure shall be repeated ten times at one minute intervals.

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4.6.2.3.2 Load switching. Generating system (with voltage set at 28 ± 0.2 volts) shall be operated at 6000 rpm when connected in a circuit as Shown in figure 1, during 150 cycles of following test procedure:

(1) Close switch 1.
(2) Close switch 3 until system voltage reaches “set voltage”.
(3) Close switch 2 to apply a 60 amp resistance load.
(4) Open switch 3 after load is applied.
(5) Open switch 2 while generating system is producing 60 amps.
(6) Wait until voltage decreases to within 10 percent of “set voltage”.

The test shall be accomplished in not more than 15 minutes at each of the following temperatures:

a. -65° ± 5°F.
b. +77° ± 15°F.

4.6.3 Current tests. To determine conformance to 3.4.2, generating system shall undergo procedures specfied in 4.6.3.1 through 4.6.3.3.

4.6.3.1 Rated output test. To determine conformance to 3.4.3.1, unless otherwise specified in applicable drawing, regulator voltage sensing lead shall be connected, during output test, to same point at which voltage is measured . Following procedures shall be observed:

a. Connect generating system in circuit as shown in figure 1.
b. Unless otherwise specified herein, battery shall remain connected for all tests and switch SW3 shall be closed.
c. Speed, frame temperature, ignition lead current, output current, and system voltage shall be recorded.
d. Voltage adjustment screw shall be rotated to stop on high voltage side of adjustment range.
e. Generating system speed and load shall be adjusted to maximum load for each point on speed-load graph represented by chart shown on applicable drawing with voltage maintained at 28 ± 0.2 volts.
f. Temperature of generating system shall be stabilized (see 4.3.2) at each point on performance curve represented by chart shown on applicable drawing.

4.6.3.1.1 Rated output - acceptance test. All generating systems furnished under this specification shall be connected into circuit shown in figure 1 with battery. Speed shall be adjusted to not more than 1800 rpm.

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Load shall be adjusted to not less than 60 amperes- Generating system shall be so operated for not less than one minute and shall give evidence of conformance to 3.4.2.1. Stabilized frame temperature will not be required.

4.6.3.2 Reduced output test. To determine conformance to 3.4.2.2, electrical adjustments shall be made as specified in drawing 10929868 and generating system subjected to test specified in 4.6.2, except maximum load shall be 25 amps.

4.6.3.3 Non-operating current test. To determine conformance to 3.4.2.3, generating system shall be connected in circuit specified in figure 1. With switches SW3 and SW1 closed, and SW2 open, non-operating steady state current shall be measured by ammeter A2. Subsequently, switch SW1 shall be opened and current measured by ammeter A1.

4.6.4 Endurance test. To determine conformance to 3.4.3, generating system shall be connected in circuit as specified in figure 1 and tested in accordance with following procedure:

a. Set regulator at 28 ± 0.2 voltage (set voltage).
b. Adjust load for 60 amps.
c. Operate generating system for a total of 1000 hours in accordance with speed and duration indicated in table VI, recording frame temperature, generating system voltage and generating system current.

TABLE VI. Endurance test - sequence.

RPM	Duration	Generating system position
2500	336 Hours	Shaft horizontal
10000	166 Hours	Shaft horizontal
6000	166 Hours	Shaft horizontal 1/
6000	166 Hours	Shaft 45° ± 1° above horizontal
6000	166 Hours	Shaft 45° ± 1° below horizontal

Sequence indicated in table VI shall be conducted three times for a total test time of 3000 hours.

NOTE: Throughout endurance test, belt loading shall be maintained in accordance with formula on applicable drawing.

1/ Throughout this test period, load shall be cycled from 0-5 amp to 60 amp (30 minutes at each load).

4.6.4.1 Endurance: reduced level first article test. When reduced level first article test is authorized (see 6.2), sequence specified in table VI shall be conducted once only for a total test time of 1000 hours.

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4.6.5 Overspeed test. To determine conformance to 3.4.4, generating system shall be connected in a circuit as specified in figure 1. With 60 amp load, generating system shall be operated at 10,000 rpm until frame temperature is stabilized (see 4.3.2.1). Subsequently, load shall be reduced to less than five amps and speed increased to 12,000 rpm. Speed and load shall continue for 15 seconds.

4.6.6 High temperature test. To determine conformance to 3.4.5, generating system shall be connected in circuit as specified in figure 1 in ambient air temperature of 250°F. With maximum load attainable at 28 Vdc + 0.2 Vdc and 1750 rpm ± 50 rpm, generating system shall be operated until frame temperature is stabilized (see 4.3.2.1). Subsequently, load shall be adjusted to 5 amps less than maximum load at which system must function continuously to maintain output voltage. This load shall be not less than 39 amps. Generating system shall be so operated for 200 hours in ambient air temperature of 250°F. Air movement in high temperature chamber shall be held to minimum to prevent temperature gradient within chamber and to prevent cooling from excessive air motion. Air motion from generating system shall not be considered in evaluating conformance. Subsequently, generating system shall undergo test procedure specified in 4.6.3 in ambient air temperature specified in 4.3.3.

4.6.7 Low temperature test. To determine conformance to 3.4.6, generating system shall undergo following procedure:

a. Prior to testing, generating system shall be stored at temperature of -80°F for a period of 24 hours.
b. After removal from storage, generating system frame temperature shall be stabilized in chamber with ambient temperature of -65°F.
c. Generating system shall be operated at 1750 ± 50 rpm with 60 amp load for a period of four hours.
d. During test, voltage shall remain at set voltage ± 0.5 volts.

Subsequently, system shall undergo test procedure specified in 4.6.3 in ambient air temperature specified in 4.3.3.

4.6.8 Shock. To determine conformance to 3.4.7, generating system shall be tested in accordance with test condition I, method 213, shock test of MIL-STD-202. Test procedure involving oscilloscope monitoring of shock impulse may be used, provided that above minimum requirements are met and method is approved by the Government. Subsequently, system shall undergo test procedures specified in 4.6.2 and 4.6.3.

4.6.9 Vibration. To determine conformance to 3.4.8, generating system shall be tested in accordance with test condition D of method 204 of MIL-STD-202. Following conditions shall apply:

a. Frequency range shall be 10 to 600 cycles per second (Hz).
b. Maximum load along longitudinal (shaft) axis shall be 10G applied at vibration table.

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c. Alternator shaft shall be rotated throughout vibration test at speed of 1750 ± 50 rpm.
d. Alternator shall be rigidly mounted as in normal vehicle operating position.

Subsequently, system shall undergo test procedures specified in 4.6.4.

466.10 Submersion. To determine conformance to 3.4.9, generating system shall be frame temperature stabilized (see 4.3.2.1) by operating at 28 volts, 2500 ± 50 rpm and load current of 60 amps. Stabilization shall be accomplished in dry condition. Generating system shall then be subjected to following test cycle:

a. Generating system shall be driven in dry condition for a period of 60 hours at 7000 ± 100 rpm and no load.
b. Alternator shaft shall then be rotated for a period of eight hours while completely submerged in a 5 percent (by weight) salt solution at a speed of 1750 + 50 rpm. At end of first hour only, rotation shall be stopped and battery current measured-while unit remains submerged. Current from battery shall not exceed 10 amperes. Switches SW1 and SW3 shall be closed and switch SW2 open for a and for first part only of b. During last seven hours of b, all switches shall be open.
c. Subsequent to submersion, regulator compartment shall be opened to determine whether solution is present and amount of solution found therein.
d. Generating system shall undergo test procedures specified in 4.6.2 and 4.6.3.

4.6.11 Fungus resistance. To determine conformance to 3.4.10, system shall be tested in accordance with method 508 of MIL-STD-810 except that test period shall be a continuous 90 days. Upon removal from cabinet, examination shall be accomplished without disturbing any fungus growth. Upon completion of examination and replacement of access port covers, generating system shall be operated for one hour without load at 6500 rpm. Subsequently, generating system shall undergo test procedures specified in 4.6.2 and 4.6.3.

4.6.12 Corrosion resistance. To determine conformance to 3.4.11, generating system shall be tested for resistance to corrosion in accordance with method 101 of MIL-STD-202, except length of test shall be 200 hours. During test, generating system will be rotated for a ten minute period out of each 8 hour test interval. After test, alternator shaft shall be rotated for 10 minutes at 1750 + 50 rpm and no load. Subsequently, generating system shall undergo test procedures specified in 4.6.2 and 4.6.3.

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4.6.13 Sand and dust test. To determine conformance to 3.4.3.9, generating system shall be subjected to method 510, sand and dust test of MIL-STD-810. Alternator position in dust chamber shall be as in intended use. Dust velocity shall be 2300 + 500 feet per minute (fpm). During 10 minutes of each hour of test period, system shall be driven at a speed of 3450 + 50 rpm.

4.7 Inspection of preparation for delivery.

4.7.1 Materials and processes. The Government inspector shall, at unscheduled intervals, inspect all materials and processes involved in preparation for delivery to determine conformance to requirements of Section 5 and specifications referenced herein. Any evidence of deviation from specified requirements shall be cause for refusal to conduct further inspection until objective evidence of corrective action has been provided by the contractor, and approved by the Government.

5. PACKAGING

5.1 Preservation, packaging, and packing. Preservation, packaging, and packing shall be in accordance with applicable packaging data sheets.

6. NOTES

6.1 Intended use. Generating systems covered by this specification are intended for use in military vehicles, and in other military equipment, to produce direct current electrical energy for various requirements, such as lighting and battery charging, under all conditions of climate, terrain, and military service.

6.2 Ordering data. Procurement documents should specify the following:

a. Title, number, and date of this specification.
b. First article testing (see 4.4).
c. Reduced level first article testing (see 4.4.1).
do Applicable drawing (see 2.1).
e. Selection of applicable level of packaging and packing (see 5.1).

6.3 First article. First article samples shall be tested and approved under the appropriate provisions of 7-104.55 of the Defense Acquisition Regulation. The contracting officer should include specific instructions in all procurement instruments regarding arrangements for examination, tests, and approval of the first article (see 3.1.1).

6.3.1 Reduced level first article inspection. Requests for reduced level first article inspection may be granted on sequential contracts where product and contractor have received first article approval under preceding contract with the Government. Such requests will not be granted where revision of

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applicable drawings, specifications, or standards adversely affects prior approvals. Applications for reduced level first article inspection should be addressed to US Army Tank-Automotive Command, ATTN: DRSTA-GB, Warren, Michigan 48090.

6.4 Change in classification. Since “class A“ and “class B“ alternators are identical and stocked under the same NSN, the classification has been dropped (see 1.2).

6.5 Recycled materials. The use of recycled materials which meet the requirements of the applicable material specifications without jeopardizing the intended use of the item shall be encouraged (see 3.2).

6.6 Definition.

6.6.1 Set voltage. As used in this specification, “set voltage” is defined as the output voltage to which generator system is adjusted.

6.7 Changes from previous issue. Asterisks are not used in this revision, to identify changes with respect to the previous issue, due to the extensiveness of the changes.

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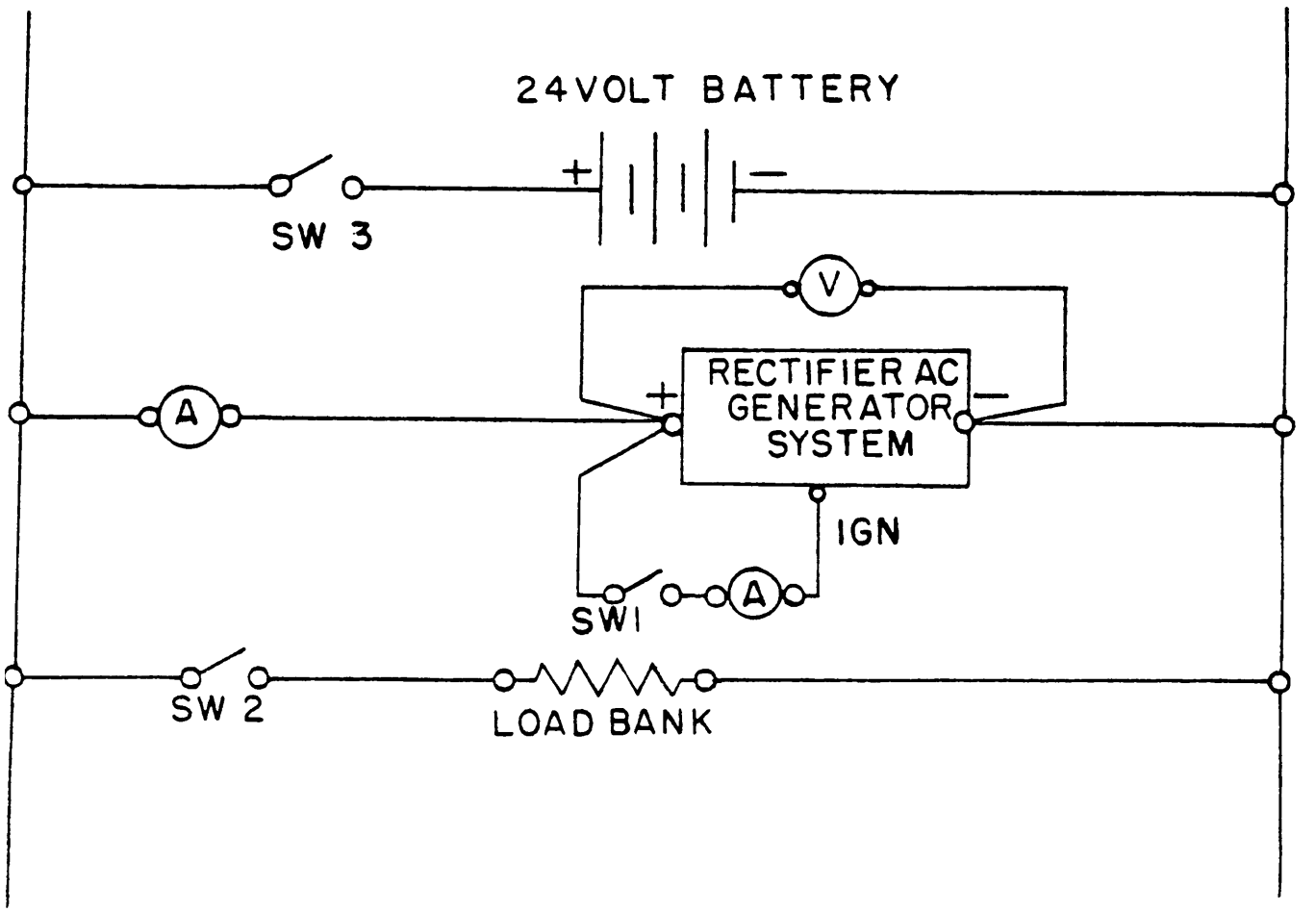


FIGURE 1 — GENERATOR SYSTEM CONNECTED IN CIRCUIT

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NEGATIVE TRANSIENT VOLTAGE TEST CIRCUIT

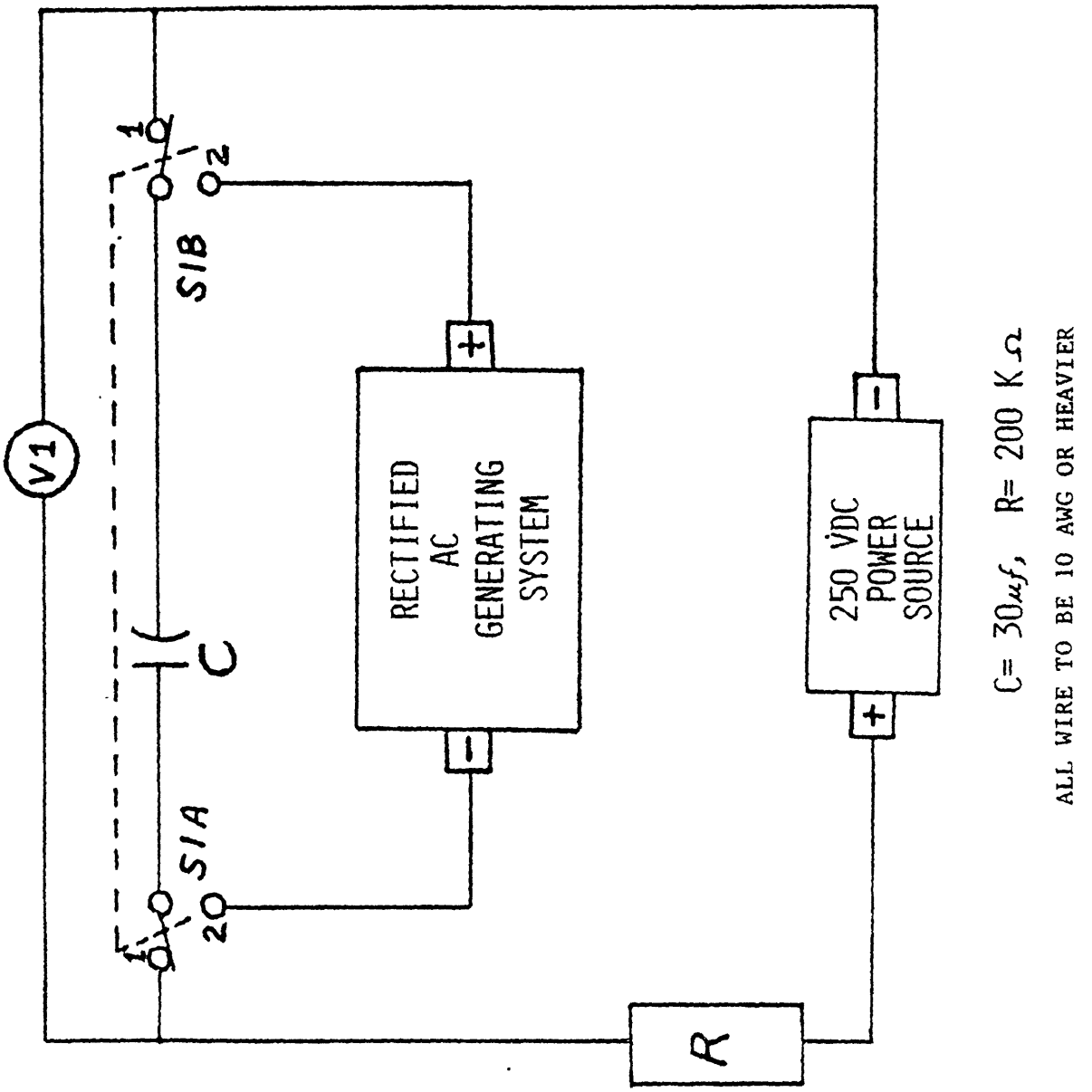


FIGURE 2

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